

Chapter 0 Exercises

0.1

A

Recall that for any random variable X we have $\mathbb{E}[(X - \mathbb{E}X)^2] = \mathbb{E}[X^2 - 2X\mathbb{E}X + (\mathbb{E}X)^2] = \mathbb{E}X^2 - 2(\mathbb{E}X)^2 + (\mathbb{E}X)^2 = \mathbb{E}[X^2] - \mathbb{E}[X]^2$. A similar linearity of expectation argument works for vectors using that $\|x\|_2^2 = x \cdot x$ with known properties of the dot product.

$$\begin{aligned}\mathbb{E}\|Z - \mathbb{E}Z\|_2^2 &= \mathbb{E}[(Z - \mathbb{E}Z) \cdot (Z - \mathbb{E}Z)] = \mathbb{E}[\|Z\|_2^2 - 2(Z \cdot \mathbb{E}Z) + \|\mathbb{E}Z\|_2^2] \\ &= \mathbb{E}\|Z\|_2^2 - 2\mathbb{E}[Z \cdot \mathbb{E}Z] + \|\mathbb{E}Z\|_2^2 = \mathbb{E}\|Z\|_2^2 - \|\mathbb{E}Z\|_2^2\end{aligned}$$

B

Recall that for scalar random variables X, X' which are independent and equal in distribution we have $\mathbb{E}[(X - X')^2] = \mathbb{E}[X^2 - 2XX' + X'^2] = \mathbb{E}X^2 - (2$